

Worked Answer Key

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1. Divide a rectangle into 8 equal parts and shade 3; numerator 3, denominator 8. 2. Eaten $\frac{5}{6}$ (numerator 5, denominator 6); left $\frac{1}{6}$ (numerator 1, denominator 6), since $6 - 5 = 1$ piece remains.

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1. Divide 0–1 into fifths; $\frac{2}{5}$ is at the second mark and $\frac{4}{5}$ at the fourth. 2. $\frac{7}{8}$ is closer to 1 because it is $\frac{1}{8}$ away; $\frac{5}{8}$ is $\frac{3}{8}$ away.

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1. $\frac{3}{7} < \frac{5}{7}$ because $3 < 5$ with equal denominators. 2. $\frac{9}{10} > \frac{6}{10}$ because $9 > 6$.

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1. $\frac{4}{5} > \frac{4}{9}$; fifths are larger pieces than ninths of the same whole. 2. $\frac{7}{12} < \frac{7}{8}$; with seven selected parts of the same whole, eighths are larger pieces. Model: draw two equal-size rectangles. Divide the first into 12 equal strips and shade 7; divide the second into 8 equal strips and shade 7. The second rectangle has more shaded area.

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1. $\frac{3}{4} = \frac{15}{20}$ (multiply by 5). 2. $\frac{18}{24} = \frac{3}{4}$ (divide both by 6).

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1. $12/18 = 2/3$ (divide by 6). 2. $35/50 = 7/10$ (divide by 5).

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1. $7/12 > 5/12$; $5/12 < 1/2$ and $7/12 > 1/2$. 2. $9/10$ is $1/10$ from 1; $5/6$ is $1/6$ away, so $9/10$ is closer.

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1. $2/3 = 10/15$ and $3/5 = 9/15$, so $2/3 > 3/5$. 2. $5/8 = 15/24$ and $7/12 = 14/24$, so $5/8 > 7/12$.

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1. $3/8 + 2/8 = 5/8$. 2. $5/12 + 4/12 = 9/12 = 3/4$.

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1. $7/9 - 2/9 = 5/9$. 2. $11/15 - 6/15 = 5/15 = 1/3$.

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1. $1/2 = 2/4$; $2/4 + 1/4 = 3/4$. 2. $2/3 = 4/6$; $4/6 + 1/6 = 5/6$.

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1. $3/4 = 6/8$; $6/8 - 1/8 = 5/8$. 2. $1/3 = 2/6$; $5/6 - 2/6 = 3/6 = 1/2$.

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1. Each whole is $5/5$, so $2\frac{3}{5} = 5/5 + 5/5 + 3/5 = 13/5$. 2. $17/4 = 4/4 + 4/4 + 4/4 + 4/4 + 1/4$: four wholes and one fourth, or $4\frac{1}{4}$.

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1. $2\frac{1}{4} + 1\frac{2}{4} = 3\frac{3}{4}$. 2. $3\frac{3}{5} + 2\frac{4}{5} = 5\frac{7}{5} = 6\frac{2}{5}$.

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1. Regroup $4\frac{1}{3} = 3\frac{4}{3}$; $3\frac{4}{3} - 2\frac{2}{3} = 1\frac{2}{3}$. 2. Regroup $6\frac{1}{4} = 5\frac{5}{4}$; $5\frac{5}{4} - 3\frac{3}{4} = 2\frac{2}{4} = 2\frac{1}{2}$.

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1. $3/5$ of 20: $20 \div 5 = 4$, $4 \times 3 = 12$. 2. $7/8$ of 32: $32 \div 8 = 4$, $4 \times 7 = 28$.

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1. $2/5 + 1/5 = 3/5$ mile. 2. $3 - 3/4 = 2\frac{1}{4}$ feet.

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1. $2/3 = 4/6$; $4/6 + 1/6 = 5/6$ cup. 2. Regroup $5\frac{1}{2} = 5\frac{2}{4} = 4\frac{6}{4}$; then $4\frac{6}{4} - 2\frac{3}{4} = 2\frac{3}{4}$ feet.

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1. No. Common denominator 12: $\frac{1}{3} + \frac{1}{4} = \frac{4}{12} + \frac{3}{12} = \frac{7}{12}$. 2. No. $\frac{5}{8} = \frac{15}{24}$ and $\frac{3}{4} = \frac{18}{24}$, so $\frac{5}{8} < \frac{3}{4}$.

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1. Use denominator 24: $\frac{5}{8} = \frac{15}{24}$, $\frac{2}{3} = \frac{16}{24}$, and $\frac{3}{4} = \frac{18}{24}$. Order: $\frac{5}{8}, \frac{2}{3}, \frac{3}{4}$. 2. $1\frac{1}{2} + 2\frac{3}{4} = 1\frac{2}{4} + 2\frac{3}{4} = 4\frac{1}{4}$ books.